RAG Assignment: Baseline vs. Re-Ranking

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1 Introduction

- Objective: compare the retrieval using a bi-encoder with retrieval followed by cross-encoder re-ranking.
- Document chosen: FIFA Club World Cup 2025 Regulations.

2 Architecture Comparison

2.1 Bi-encoder

The bi-encoder encodes the question and each document chunk indipendently into a fixed-size vecotrs. The document embeddings are computed once and they are stored in the FAISS index formate. During the query time, new embedding is computed for the question only. Relevance is estimated using cosine similaity:

$$s_{\mathrm{BI}}(q, d) = \frac{E(q)^{\top} E(d)}{||E(q)||||E(d)||} \tag{1}$$

In the current implementation, embeddings are normalized, so FAISS inner-product search gives cosine similarity, this supports fast retrieval, but the question and passage do not directly interact during encoding. Similar passages can receive high scores even when they omit a decisive condition.

2.2 Cross-encoder

The cross-encoder processes the question and a document chunk together as a single input sequence. Joint self-attention allows the question tokens and passage tokens to interact during encoding. The model produces a relevance score for each question–passage pair:

$$s_{\mathrm{CE}}(q, d) = f_{\mathrm{CE}}([q; d]) \tag{2}$$

Here, $[q; d]$ represents the question and passage combined with special separator tokens. Unlike cosine similarity, this score is directly predicted by the model.

In the current implementation, the cross-encoder scores the same ten chunks retrieved by the bi-encoder and reorders them by decreasing relevance score. The top three chunks are then passed to the language model. Joint processing can help identify passages containing relevant conditions or exceptions, but it requires a separate model computation for each question–passage pair, which adds retrieval latency.

3 Qualitative and Rank Shift Analysis

Both pipelines pass the top three chunks to the same language model using the same prompt template. Answer 1 uses the baseline ranking, while Answer 2 uses the reranked results. The answer comparisons below are summaries of the generated outputs.

3.1 Provisional Squad Goalkeeper Requirement

Query: We use only three goalkeepers in our final tournament squad. Is listing those same three enough for the earlier provisional player submission?

Chunk `C0066`, containing Article 23.1 on PDF page 32, moved from rank 4 to rank 3. This article states that the provisional list must contain at least four goalkeepers.

In Answer 1, the model could not determine whether three goalkeepers were sufficient because the required clause was outside the top three chunks. In Answer 2, the model correctly answered no and explained the difference between the provisional requirement of four goalkeepers and the final-list requirement of three. The baseline failure was an abstention caused by missing essential evidence, rather than a hallucination.

3.2 Protests After the Final Match

Query: Does the usual two-hour deadline still allow a protest once the final match of the competition has ended?

Chunk `C0030`, containing Article 8.8 on PDF page 14, moved from rank 5 to rank 3. This clause states that protests covered by Article 8 are disregarded after the final match ends.

In Answer 1, the model could not determine the answer and discussed general protest procedures without the final-match cutoff. In Answer 2, the model correctly answered no by applying this cutoff, while noting FIFA's separate power to initiate disciplinary proceedings. Reranking recovered the specific restriction needed to answer the question.

3.3 Regression and Remaining Failure

Reranking did not improve every answer. For Q36, which asks whether provisional squad lists are published, chunk `C0067` moved from rank 2 to rank 5. Article 23.4 on PDF page 32 states that these lists are not published. Answer 1 was correct, but Answer 2 abstained after losing this evidence.

For Q25, another question about protests after the final, chunk `C0030` remained at rank 4 in both pipelines. Answer 1 incorrectly suggested that the protest could still be accepted under ordinary procedures, while Answer 2 also failed to apply the final-match cutoff.

These cases show that reranking can recover essential evidence, but it can also remove it or leave it outside the top three. Even a one-position change from rank 4 to rank 3 can affect the generated answer.

4 Latency Profiling

The runtime was measured over 28 queries with five repetitions each, giving 140 measurements per stage. The models ran on an RTX 4060 Laptop GPU, while FAISS search ran on the CPU. These timings exclude model loading, index construction, and LLM answer generation.

Stage	Mean (ms)	Median (ms)	P95 (ms)
Baseline retrieval	8.88	8.08	15.01
Additional reranking	20.67	19.92	27.63
Retrieval + reranking	29.55	28.97	42.72

Table 1: Retrieval latency across 140 measurements per stage.

The reranking pipeline took approximately 3.33 times the baseline retrieval time, adding 20.67 ms on average. This additional time comes from scoring the ten question–passage pairs using the cross-encoder. Reranking improved evidence retrieval for some queries, but introduced additional computational cost.

5 Scaling to the Entire Document Collection

The cross-encoder processes the question and each document chunk together, so its computation must be repeated for every question–passage pair. Unlike the bi-encoder, it cannot reuse independent document embeddings to calculate relevance scores for new queries. Running it across the entire collection becomes expensive as the number of chunks increases.

In the current implementation, the collection contains 132 chunks. Applying the cross-encoder directly would require scoring 132 pairs per query, compared to only ten after bi-encoder retrieval. This is 13.2 times as many pairs, although it does not necessarily mean the runtime increases by the same factor.

The two-stage pipeline uses fast embedding search to select candidates and applies the more expensive cross-encoder only to these candidates. However, this approach has a limitation: if the required evidence is missing from the initial top ten chunks, reranking cannot recover it.